

Lakota Symbols Experiment — Setup Guide

Run the experiment (`lakota_symbols_triggerhub-02.psyexp`) in **PsychoPy Builder** with the **DSI-24** EEG headset and the **Trigger-Hub (MMBT)** marker box.

Two roles are involved:

- **Stimulation computer** — runs PsychoPy; the **MMBT** is plugged into it.
- **Recording computer** — runs DSI-Streamer; the **DSI Bluetooth dongle** is plugged into it.

These can be **two computers or one**. On a single computer the same machine plays both roles — just do every step on it.

Triggers travel **wirelessly** from the MMBT to a small **receiver dongle plugged into the DSI-24**, which writes the marker straight into the EEG. The computers never need to be networked — the marker is recorded inside the EEG itself.

1. What you need

Item	Where it goes	Notes
DSI-24 headset	—	WearableSensing dry EEG
Trigger receiver dongle	in the DSI-24	receives triggers wirelessly from the MMBT
DSI Bluetooth dongle	recording PC (USB)	the DSI-24 streams EEG to it; replaces the PC's built-in Bluetooth
MMBT (Trigger-Hub) box	stimulation PC (USB)	shows up as " Arduino Micro "
PsychoPy (Builder)	stimulation PC	runs the experiment
DSI-Streamer	recording PC	records the EEG — <code>DSI-streamer-and-drivers/DSI-Streamer-v.1.08.125_Released/DSI-Streamer-v.1.08.125.exe</code>
Tachyon	stimulation PC	quick trigger test — installer at <code>DSI-streamer-and-drivers/mmbt/TachyonSetup.exe</code>

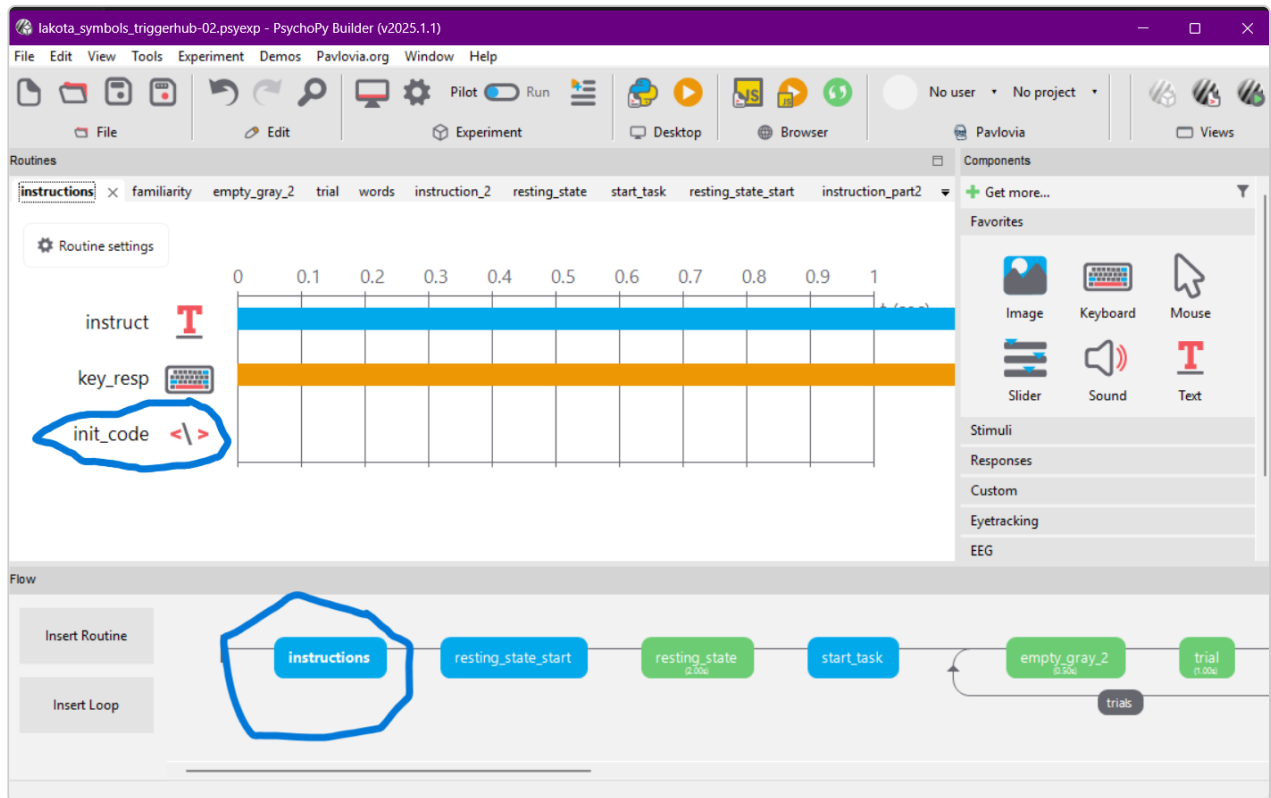
MMBT drivers and guides are in `mmbt/` .

2. One-time setup

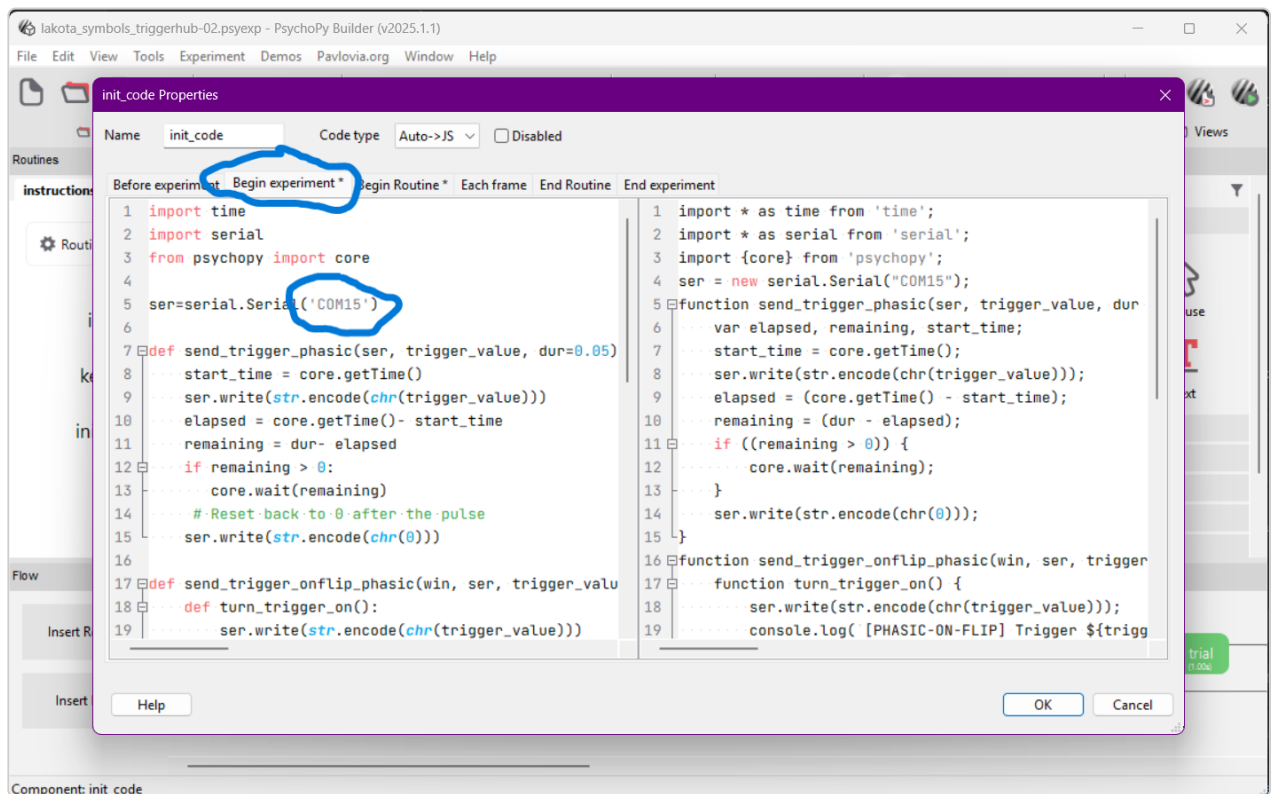
Stimulation computer

1. Install **PsychoPy** (Standalone, 2025.1.x) — <https://www.psychopy.org/download.html>. `pyserial` is bundled, nothing else to install.
2. Plug in the MMBT and install its driver from `mmbt/MMBT drivers/` : run `dpinst-amd64.exe` (or `dpinst-x86.exe` on 32-bit). It then appears as **Arduino Micro** under Device Manager → **Ports (COM & LPT)** — note its COM number.
3. **Set the COM port in the experiment** (in Builder):

Open `lakota_symbols_triggerhub-02.psyexp` in **PsychoPy Builder**, and in the **instructions** routine click the `init_code` component:



Then open the **Begin Experiment** tab and set your COM number in the line `ser = serial.Serial('COM15')`:



Save. *Tip:* pin the COM number in Device Manager → port **Properties** → **Port Settings** → **Advanced** → **COM Port Number** so it doesn't drift.

4. Install **Tachyon** (`DSI-streamer-and-drivers/mmbt/TachyonSetup.exe`) for the trigger test.

Recording computer

1. Plug in the **DSI Bluetooth dongle**. It acts as the computer's Bluetooth for the DSI — use it, **not** the PC's built-in Bluetooth.
 2. Power on the DSI-24 and **pair** it with the dongle (Windows **Bluetooth settings** → **Add device** → select the DSI).
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3. Steps to run the experiment

Connect the hardware

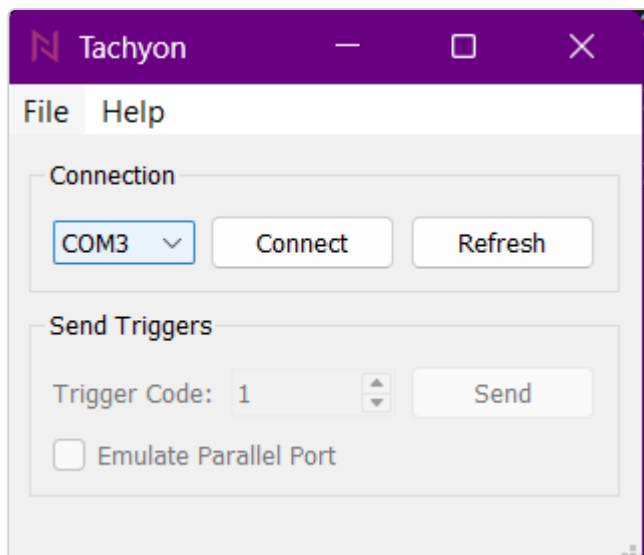
1. **Stimulation PC:** plug in the **MMBT** (USB).
2. **DSI-24:** plug the **trigger receiver dongle** into it.
3. **Recording PC:** make sure the **DSI Bluetooth dongle** is plugged in.
4. Power on the **DSI-24** and the **MMBT**.

Start the EEG (recording PC)

1. Open **DSI-Streamer** (`DSI-streamer-and-drivers/DSI-Streamer-v.1.08.125_Released/DSI-Streamer-v.1.08.125.exe`), pick the DSI's **outgoing** COM port, and press **Connect**.
Pairing the DSI creates **two** Bluetooth COM ports for it — an *incoming* and an *outgoing* one. Use the **outgoing** port (the COMxx identified as the DSI's *out* port), **not** the incoming one. Find it in **Device Manager** → **Ports (COM & LPT)**, or under Windows **Bluetooth** → **More Bluetooth options** → **COM Ports** (the entry with **Direction** = **Outgoing**), then select that COMxx in DSI-Streamer.
2. Fit the headset and check impedances (< 50 kΩ).

Test the triggers — with Tachyon, *before* recording

1. On the stimulation PC, open **Tachyon** → select the MMBT COM port (the **Arduino Micro** port) → **Connect** → set a **Trigger Code** (e.g. 1) → **Send**:



2. Watch DSI-Streamer: the trigger/event channel should jump to that value. If it stays at 0, re-check steps 1–2 and the COM port.
3. **Close Tachyon** — it holds the COM port open, and PsychoPy can't use the MMBT until it's released.

Record and run

1. In DSI-Streamer, start **recording**.

2. On the stimulation PC, open `lakota_symbols_triggerhub-02.psyexp` in **PsychoPy Builder** and click **Run** (the green running-figure ►). Enter participant / session.
3. Run the task — the MMBT marks each phase in the EEG as you go.
4. When finished, **stop recording** in DSI-Streamer.

4. Checklist

- ☐ MMBT driver installed; **Arduino Micro** shows a COM port
- ☐ COM port set in Builder (`init_code` → Begin Experiment → `serial.Serial('COMxx')`)
- ☐ Trigger receiver dongle in the DSI-24
- ☐ DSI Bluetooth dongle on the recording PC; DSI-24 powered on & paired
- ☐ DSI-Streamer **Connect**, impedances OK
- ☐ Tachyon test → trigger shows on the event channel → **Tachyon closed**
- ☐ DSI-Streamer **recording** started
- ☐ PsychoPy Builder **Run** → participant entered

5. Troubleshoot

Symptom	Fix
could not open port 'COMxx'	COM port wrong or in use — close Tachyon (or any app holding it), check Device Manager, fix it in Builder.
MMBT not listed as Arduino Micro	Driver not installed — run <code>dpinst-amd64.exe</code> in <code>mmbt/MMBT drivers/</code> .
No triggers on the event channel	MMBT on USB & powered, trigger receiver dongle seated in the DSI-24, correct COM port. Confirm with Tachyon.
DSI won't connect	Use the DSI Bluetooth dongle (not built-in BT, not the trigger dongle); headset charged/on & paired; then Connect in DSI-Streamer.
COM number keeps changing	Pin it in Device Manager → Port Settings → Advanced → COM Port Number.

For the experiment flow and the full trigger-code list, see the **README**.